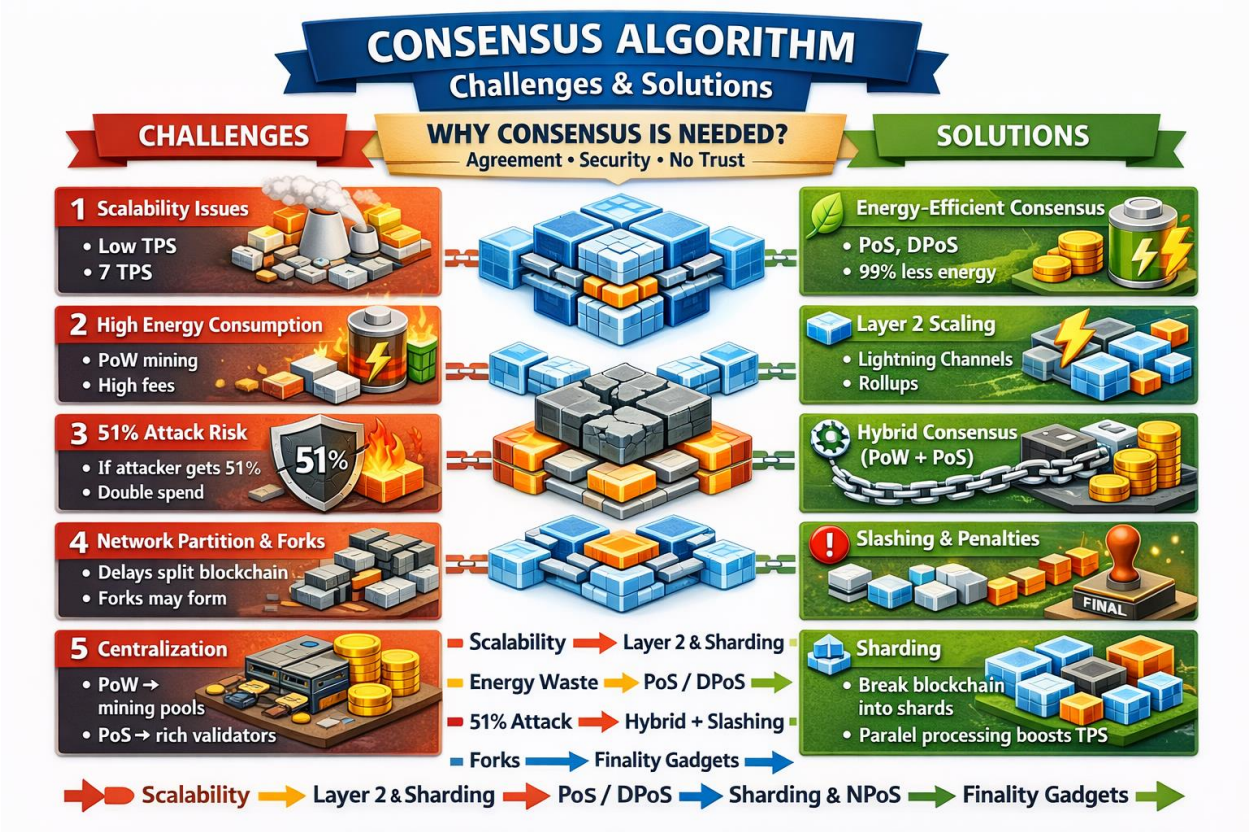
UNIT–II



MODELING FAULTS & ADVERSARIES

• Blockchain in Hostile Environments

Blockchain in Hostile Environments → Nodes Can Fail, Misbehave, or Attack

A. FAULT MODELS

1 Crash Faults
- Node goes offline due to crash, power failure

2 Omission Faults
- Message not sent/received → network delay or overload

3 Timing Faults
- Node responds too late → deadlines missed → forks possible

4 Byzantine Faults (Most Serious)
- Node behaves maliciously:
- lies
- sends fake messages
- colludes with attackers
- double spends

B. ADVERSARY MODELS

1 Honest Nodes
- Follow protocol

2 Rational Nodes
- Profit-driven, (may cheat if beneficial)

3 Malicious Nodes
- Intentionally attack blockchain

4 Weak Adversary
- Very limited resources

5 Strong Adversary
- Can delay/reorder messages, control many nodes

Weak Adversary → Adversary → Strong Adversary → Super Adversary

Byzantine Fault Tolerant (BFT) Models

Byzantine Generals Problem

How can all generals (nodes) agree when some may be traitors?

Core Rule: 3f + 1 Nodes

A blockchain tolerates f malicious nodes only if total nodes $\geq 3f+1$

1. Classical PBFT
- 3 phases (pre-prepare, prepare, commit)

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- Used in private/consortium blockchains
- Fast finality
- $O(n^2)$ communication → not scalable

2. Federated BFT
- Ripple Stellar
- Quorum slices
- Flexible trust

3. Delegated BFT (dBFT)
- NEO

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4. BFT in PoS (Ethereum, Cosmos, Polkadot)
- Validators vote
- 2/3 majority → finality
- Strong security

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